

CUSTOMER MILESTONE BRIEF

MS #42 - Integrated Pilot Qualification

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.42.0
Date	2026-08-31
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #42 - Integrated Pilot Qualification converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Proves that independently qualified COBOL, PL/I, JCL, DB2, and HLASM subsets compose into one governed customer-shaped pilot.

Why this matters to a customer

Proves that independently qualified COBOL, PL/I, JCL, DB2, and HLASM subsets compose into one governed customer-shaped pilot.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Bound the exact five-cell ACCOUNTV pilot selected in MS #32 to its six source files, five technologies, five coordination dependencies, and canonical source-graph identity.
- Added a deterministic 40-case integrated corpus covering online and batch paths, Db2 cardinality, schema and index behavior, copybook layout, JCL flow and datasets, HLASM branching, cross-language calls, source identity, dependency edges, and four fail-closed native vectors.
- Preserved material source truths: unqualified single-row SQL cardinality, the PL/I account-ID overwrite, the bounded null/non-null DATEFMT branch, and the external CBACT04C boundary.
- Added five bounded cell receipts, ten acceptance-evidence items, 15 blocked live-evidence items, a 30-entry compatibility ledger using all five governing classes,

and twelve qualification gates.

- Added frozen schemas, POSIX/PowerShell release-gate integration, deterministic artifacts, graph holdouts, source/dependency drift probes, mutation cases, and rehashed-overclaim tests.
- Marked Wave 2 integrated development evidence ready while keeping factory dispatch, native compilation/execution, mainframe equivalence, production release, and production readiness false.

How it differs from earlier work

MS #42 builds on MS #41 (HLASM Qualification Hardening). The earlier milestone established its own bounded capability; this milestone adds integrated pilot qualification while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use integrated pilot qualification as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.42.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications

- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.